SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.
Assessment Tool Weightage: 20%

QUESTIONS: 05 Total Marks:50

Annexure A

COMPARITIVE ANALYSIS

Criteria	Scope of Items ***(2)***	Comparison Depth(2.5)	Use of Evidence ***(2)***	Critical Thinking ***(2)***	Clarity of Presentation ***(1)***	***Total(10)***

Weightage	*20%*	25%	20%	20%	10%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code

of Conduct:

I _____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action. ***Signature***

of the student:

RUBRICS FOR CALCULATION:

		Comparative analysis			
Criteria	**Weight ag e (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developi ng)**	**Level 1 (Beginni ng)**
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant

Compari so n Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparis on with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretati on with minor gaps	Basic interpreta tio n; lacks depth	No meaningful interpreta tio n
Clarity of Presenta tio n	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorgani ze d and difficult to understand

Annexure A

Q1 . Dataset: Assessment Tool 3: Comparative Analysis (25% → 5 Marks)

Time duration :45 Minutes
Assessment Tool 3 (Low)

Q1 . Dataset: Customer Data

Customer Income (₹) Spending Score
C1 30000 60
C2 60000 40
C3 50000 70
C4 28000 65
C5 65000 35
Question:
Compare K-means and hierarchical clustering using this dataset and explain differences in performance and scalability. *(BL3 – 1 Mark)*

Q2. Dataset: Document Data

Document Word1 Word2 Word3
D1 0.2 0.5 0.3
D2 0.1 0.6 0.3
D3 0.8 0.1 0.1
D4 0.75 0.15 0.1

D5 0.2 0.4 0.4
Question:
Compare K-means and EM algorithms in handling overlapping clusters for the given dataset. *(BL4 – 1 Mark)*

Q3. Dataset: Geographic Data

Point	Latitude	Longitude
P1	13.08	80.27
P2	13.10	80.25
P3	12.97	80.22
P4	13.05	80.30
Point	**Latitude**	**Longitude**
P5	13.00	80.20

Question:
Compare Euclidean and Manhattan distance metrics and analyze their impact on clustering results. *(BL4 – 1 Mark)*

Q4. Dataset: Retail Products

Product	Sales (₹)	Frequency
P1	50000	20
P2	30000	10
P3	52000	22
P4	25000	8
P5	32000	12

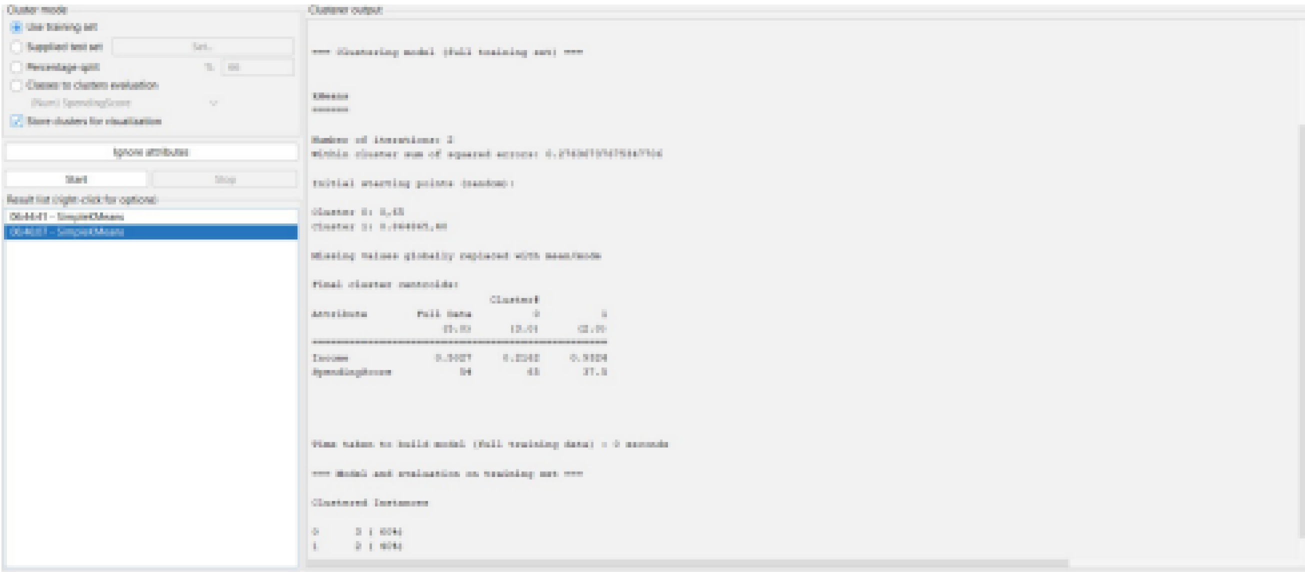
Question:
Compare agglomerative and divisive clustering approaches and explain their suitability for this dataset. *(BL3 – 1 Mark)*

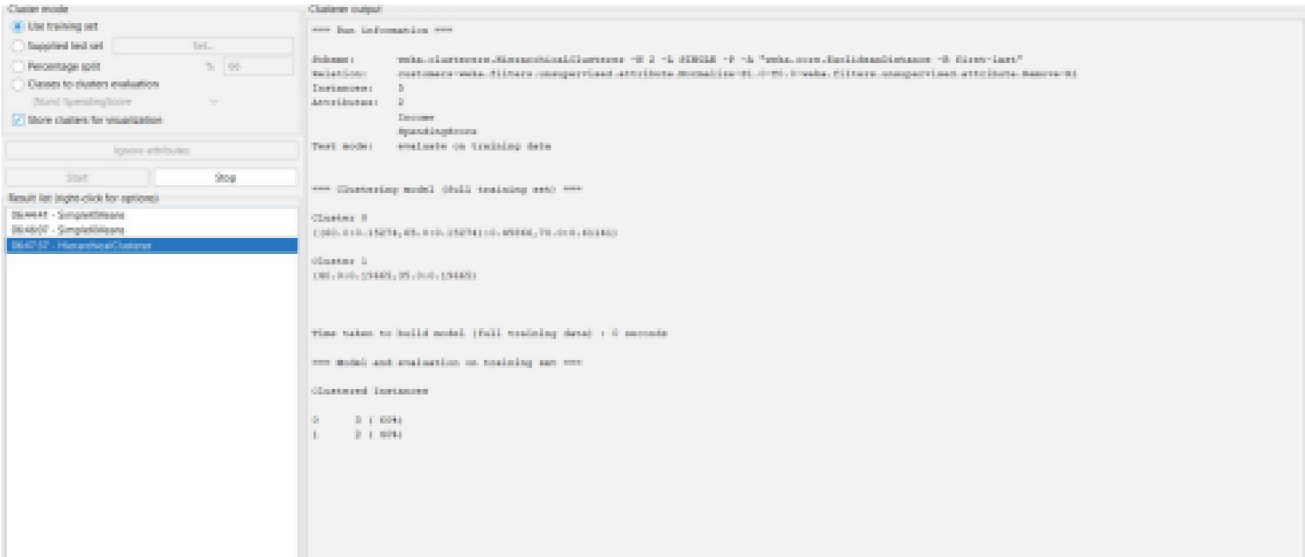
Q5 . Dataset: Mixed Attributes

Item	Price (₹)	Category	Rating
I1	500	A	4.5
I2	300	B	3.8
I3	550	A	4.7
I4	250	B	3.5
I5	520	A	4.6

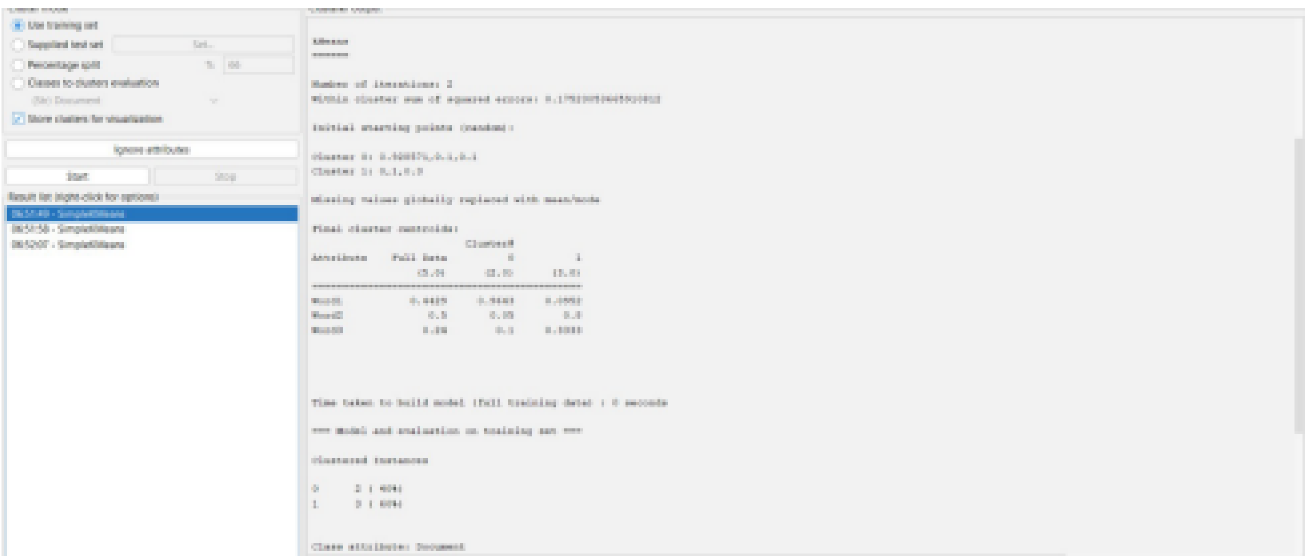
Question:
Compare centroid-based and density-based clustering techniques and analyze their effectiveness on mixed data types. *(BL4 – 1 Mark)*

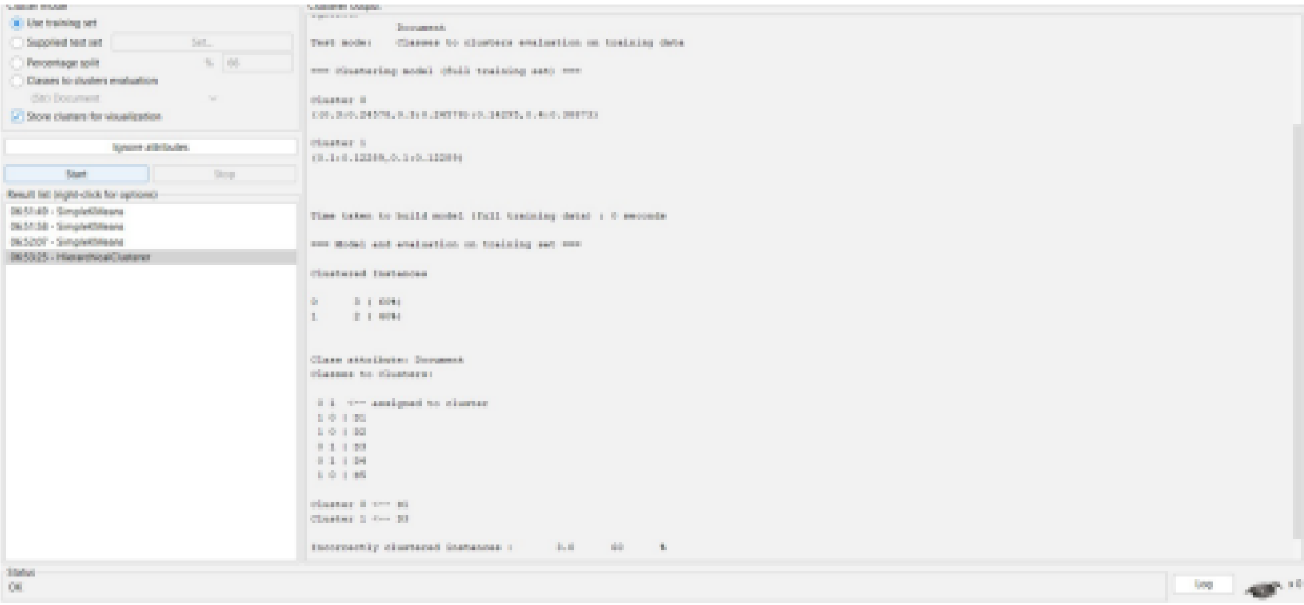
Q1 - Answer



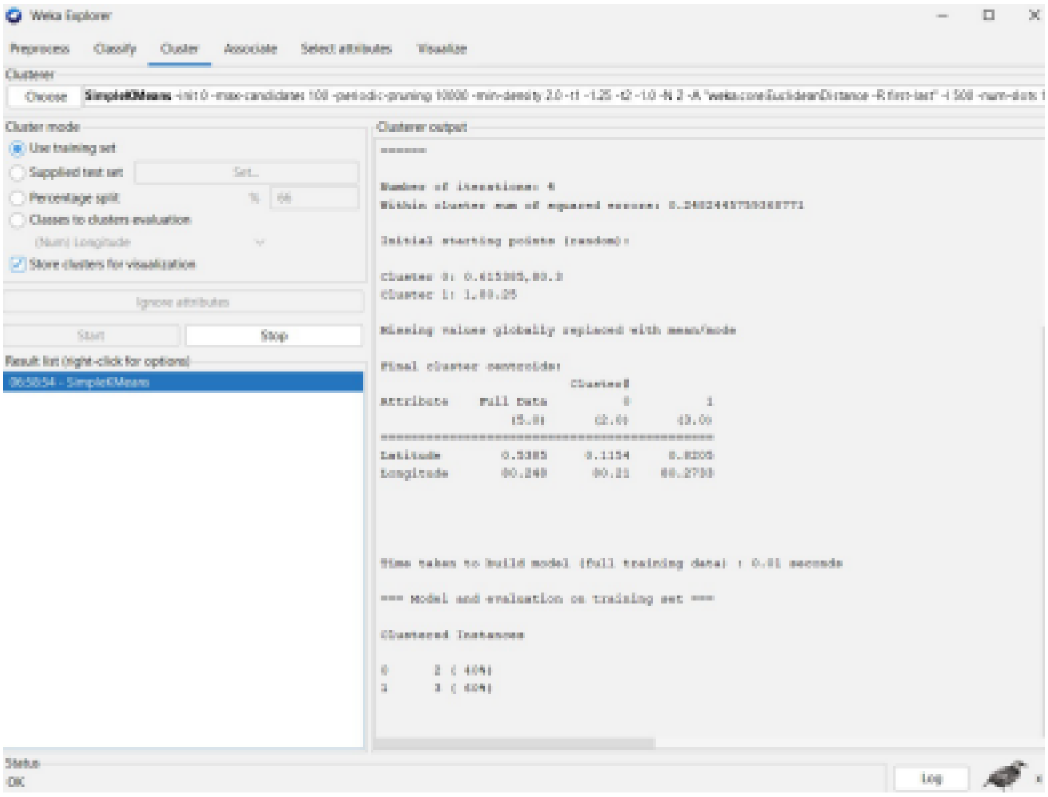


Q2 - Answer

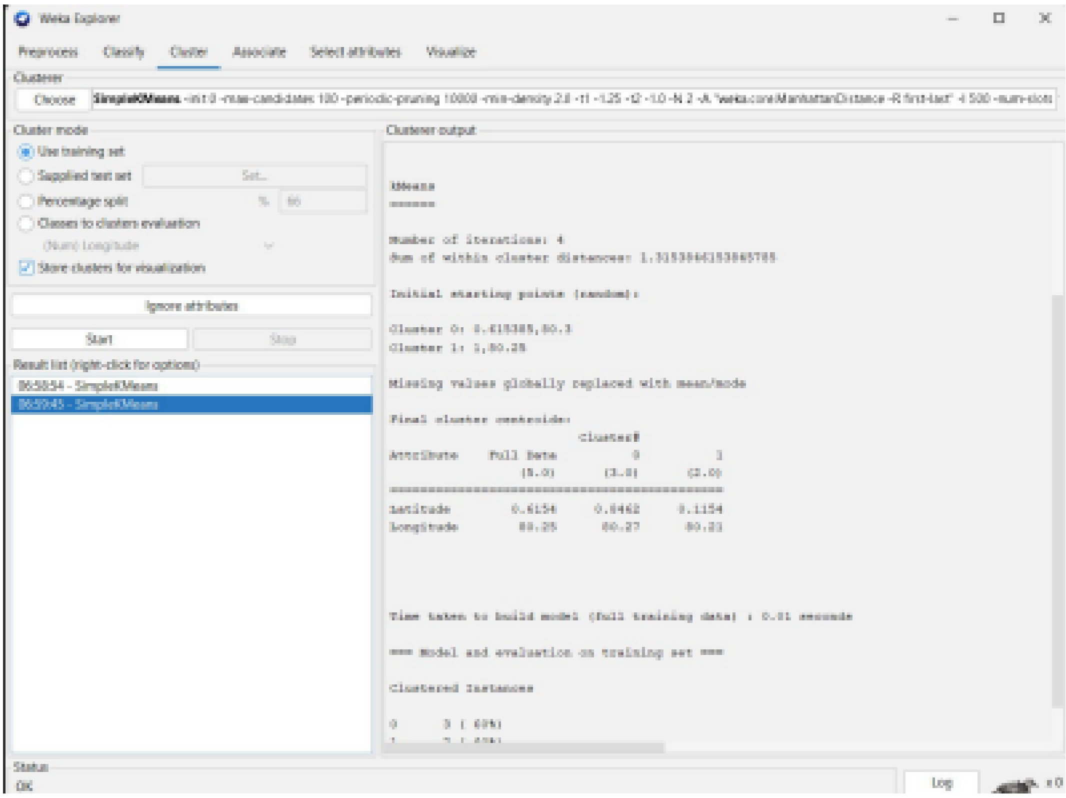




Q3 - Answer



Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
Cluster mode
Use training set
Supplied test set
Percentage split
Classes to clusters evaluation
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
Clusterer output
Status
OK
Log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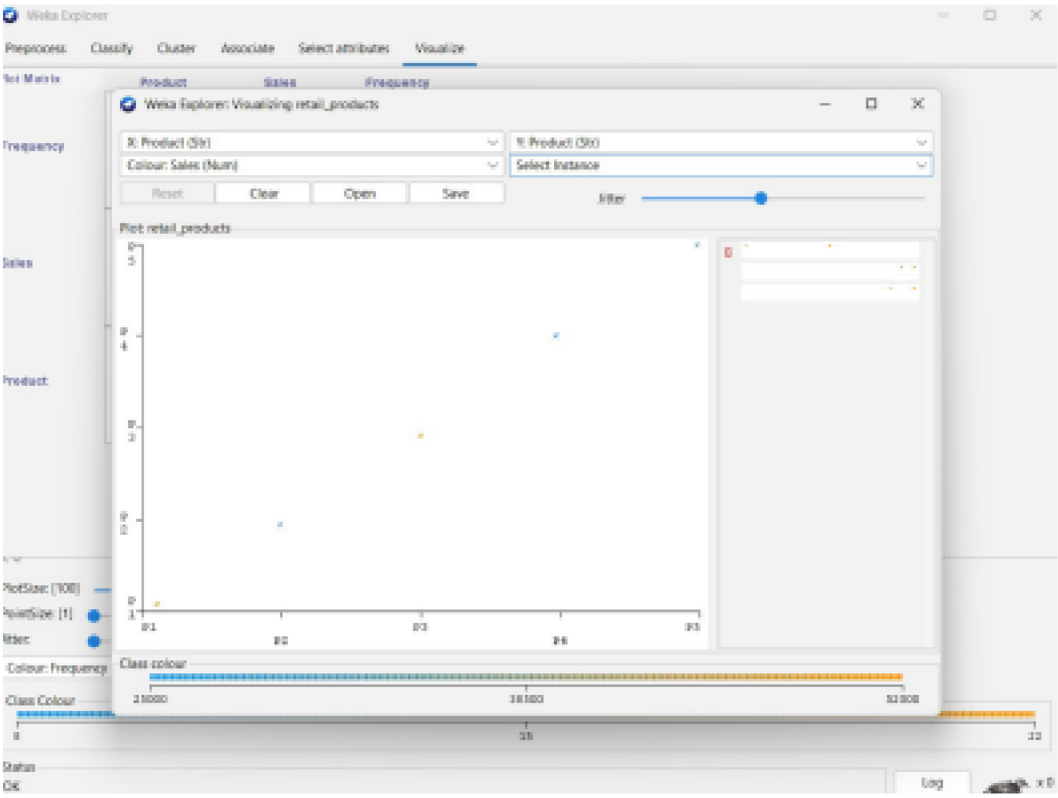
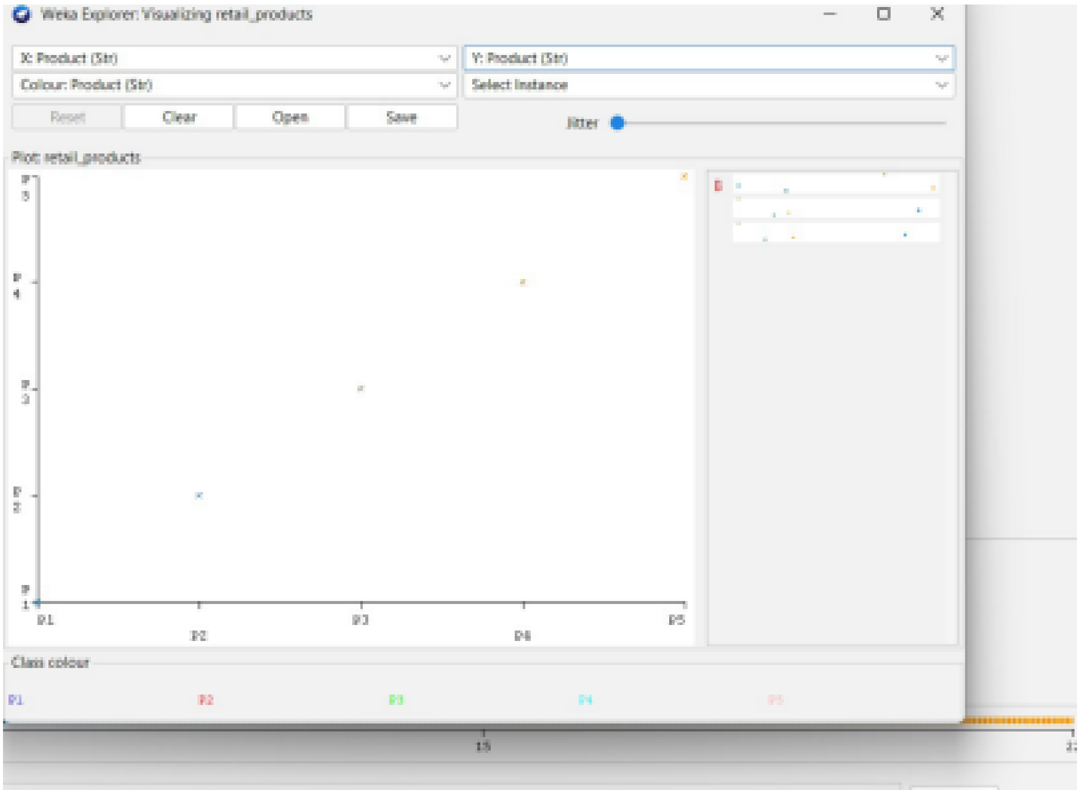
Q4 - Answer


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Weka Explorer: Visualizing retail_products
X: Product (Str)
Y: Product (Str)
Colour: Sales (Num)
Select Instance
Reset
Clear
Open
Save
Jitter
Plot: retail_products
Class colour
Status
OK
Log


Weka Explorer: Visualizing retail_products
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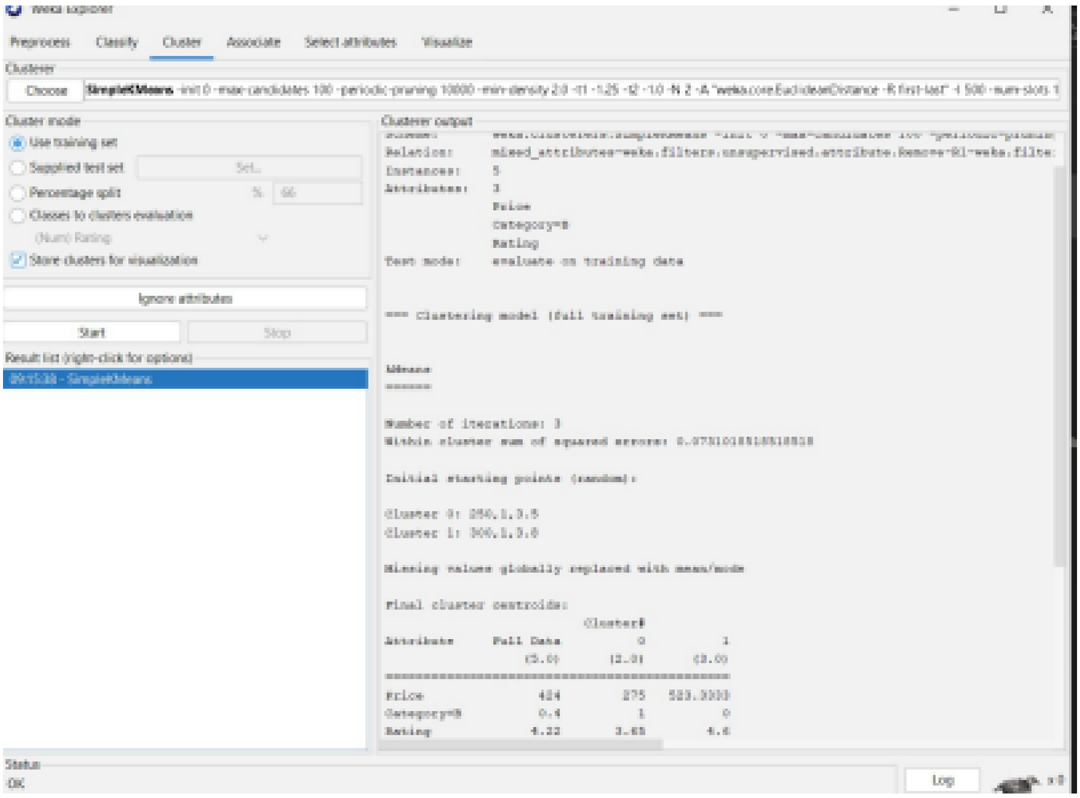
Select Instance
Reset
Clear
Open
Save
Jitter
Plot: retail_products
P1
P2
P3
P4
P5
Class colour

Q5 - Answer


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
SimpleKMeans
Cluster mode
Use training set
Supplied test set
Percentage split
Classes to clusters evaluation
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
Clusterer output
Instances: 5
Attributes: 3
Price
Category=B
Rating
Test mode: evaluate on training data
=== Clustering model (full training set) ===
kMeans
======
Number of iterations: 3
Initial starting points (random):
Missing values globally replaced with mean/mode
Final cluster centroids:
Cluster#
Attribute Full Data 0 1
(5.0) (2.0) (3.0)
Price 424 275 523.3333
Category=B 0.4 1 0
Status
OK
Log